



# United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Midterm Exam, Trimester: Summer 2025

Course Code: CSE-3521 Course Title: Database Management Systems

Total Marks: 30

Duration: 1 hour 30 minutes

**Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.**

1. a) *La Liga*, a Spanish football league consisting of multiple clubs. Each club is uniquely identified by its name and further characterized by its city, stadium capacity, and annual revenue. Clubs are associated with professional football players, who are identified by a player ID and have attributes such as name, position, nationality, salary, and category (star, regular, or youth academy). 7 [CO1]  
  
Players sign contracts with clubs, and each contract contains details such as a contract ID, start date, end date, and agreed salary. Clubs make installment-based payments under these contracts, with each payment identified by a payment number, payment date, and amount.  
  
Matches are played between two clubs and are identified by a match ID, along with attributes like date, venue, and scoreline. Players participate in matches, where their minutes played and goals scored are recorded.  
  
In addition to players, each club employs staff members such as coaches and medical personnel, identified by a Staff ID, name, and role. Staff may report to a head coach and can also have dependents (e.g., spouse, children).  
  
Now, **draw an ER diagram** for the scenario capturing all the entities, their cardinalities and relationship.
- b) In a relational database, consider these 2 tables: the **Orders** (OrderId[pk], Customer\_id[fk], amount) and the **Customers** ( CustomerID[pk],name,salary) table. 2 [CO1]
  - What would happen if you try to insert an order with a **CustomerID** that does not exist in the **Customers** table? **Explain** briefly if any integrity rule is violated.
2. a) **Draw** the corresponding Schema diagram based on the following ER diagram of a Social Media Database provided in Fig 1, 6 [CO1]

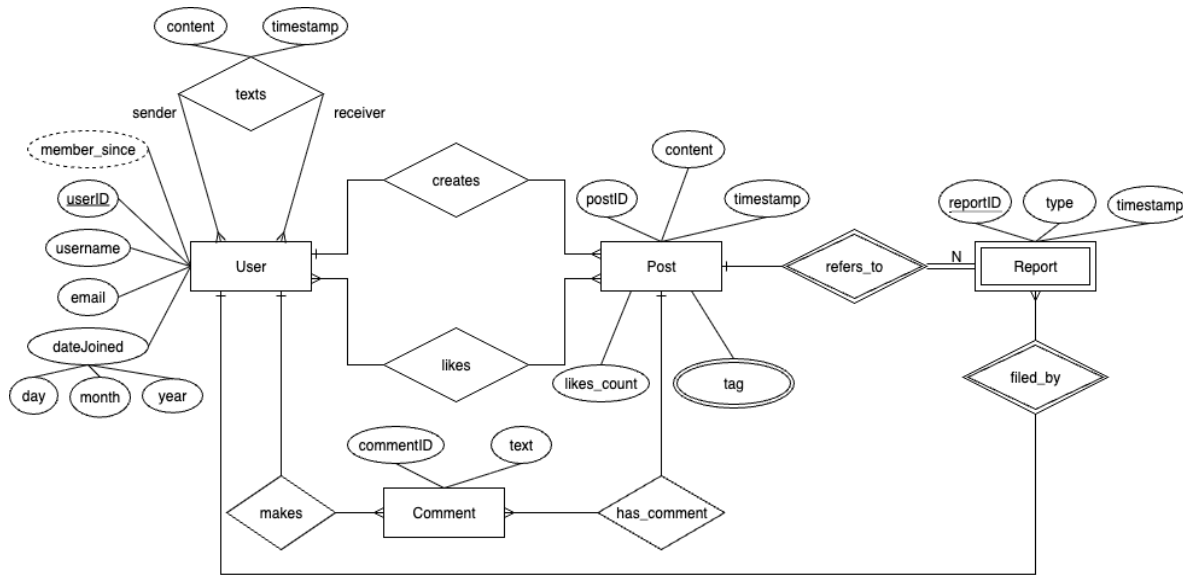


Fig 1, ER diagram of a Social Media Database

1  
[CO1]

- b) What characteristics make a candidate key a good choice for a primary key? Using a practical example with two candidate keys, **explain** how you would **choose** one and **justify** your decision.

3. Consider the following schema,

1 + 1 + 2 = 4  
[CO2]

**Movie**(movie\_id (pk), title, genre, release\_year, director\_id (fk), duration\_minutes)

**Director**(director\_id (pk), name, birth\_year, nationality)

**Actor**(actor\_id (pk), name, birth\_year, nationality)

**StarsIn**(actor\_id (fk)(pk), movie\_id (fk)(pk))

**Review**(review\_id (pk), movie\_id (fk), rating, review\_text)

Answer the following questions using **Relational Algebra**:

- Find** the average rating for each movie
- List** the names of actors who have not starred in any *Thriller* movie
- Find** the titles of movies that are directed by either *Christopher Nolan* or *Steven Spielberg*

4. Suppose an already created database schema is given below having following tables,

5 \* 2 = 10  
[CO2]

**Branch**(branch\_id(pk), name, city, rating, manager\_id[fk])

**Librarian**(librarian\_id(pk), name(pk), role, salary, branch\_id[fk])

**Member**(member\_id(pk), name, email, phone, address)

**Book**(book\_id(pk), title, genre, price)

**Section**(section\_id(pk), branch\_id[fk], category, floor, status, max\_capacity)

**Loan**(loan\_id(pk), member\_id[fk], book\_id[fk], branch\_id[fk], loan\_date, return\_date)

**Service**(service\_id(pk), section\_id[fk], service\_type, cost)

Now write **SQL** queries for the following,

- a) **Write** an SQL statement to add foreign key to the **Service** table from the **Librarian** table.
- b) **Alter** the **Section** table to change the data type of section\_id from **String** to **Integer** ensuring the structural consistency considering all integrity constraints.

Write down these following **SQL** considering the changes made in question 4(a) and 4(b),

- c) **Find** the **names and salaries** of librarians who earn **less than 30,000** but the name should not end with any vowel or shouldn't have quotation at the beginning or end of the name.
- d) **Find** the **members** who have borrowed books from branches located in '**Dhaka**' after **2024-01-01**.
- e) **Identify librarians** who have provided **service** to sections of branches where the branch rating is not less than 4. Display the **name** and **role** of the librarian including the branch name.